Composing Piecewise and Nonlinear Functions
Answer Key
Precalculus

Quick Answers

1. 101
2. $2x^2 - 1$
3. -3, 11
4. (a) value = 117, —
5. $\frac{2x}{1-x}$
6. 11
7. (a) f(g(-5)) = 13, (b) g(f(-5)) = 17
8. $x^2 - 9$
9. 1, 3
10. (a) value = 18, —
11. $\frac{1}{x-1}$
12. (a) x = -1, 4, —

What is verified

13 of 16 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 4, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSF-BF.A.1c – *problems 2, 3, 5, 6, 8, 9, 11, 12*
HSF-IF.A – *problems 1, 4, 7, 10*
Difficulty 1–5 of 5 across 16 tagged checks.

1. $f(x) = x^2 + 1$, $g(x) = 3x - 2$. Find $(f \circ g)(4)$.
Solution: Inner function first: $g(4) = 3(4) - 2 = 10$, then $f(10) = 10^2 + 1 = 101$.

$$\boxed{(f \circ g)(4) = 101}$$

2. $p(x) = 2x + 5$, $q(x) = x^2 - 3$. Write $(p \circ q)(x)$.
Solution: Substitute $q(x)$ into every x of p: $p(q(x)) = 2\left(x^2 - 3\right) + 5 = 2x^2 - 6 + 5$.

$$\boxed{(p \circ q)(x) = 2x^2 - 1}$$

3. $u(x) = x - 4$, $v(x) = x^2$. Solve $(v \circ u)(x) = 49$.
Solution: $(v \circ u)(x) = (x-4)^2$, so

$$(x-4)^2 = 49$$
$$x - 4 = \pm 7$$

Both signs give a solution: $x = 11$ and $x = -3$.

$$\boxed{x = -3 \quad \text{or} \quad x = 11}$$

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4. $g(x) = x^2 - 5$ and f piecewise ($2x + 1$ for $x < 3$; $x^2 - 4$ for $x \geq 3$). **(a)** $(f \circ g)(4)$. **(b)** Which rule and why.

Solution (a): $g(4) = 4^2 - 5 = 11$. Since $11 \geq 3$, the second rule applies: $f(11) = 11^2 - 4 = 117$. $\boxed{(f \circ g)(4) = 117}$

Solution (b): *Instructor-judged.* The response must name the inner value 11 and compare it with the boundary 3: because $11 \geq 3$, the rule $x^2 - 4$ is the one that applies. Naming the rule without the comparison is only half the answer; using $2x + 1$ means the student tested the original input 4 instead of the value actually being fed into f.

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5. $f(x) = \dfrac{2}{x-1}$, $g(x) = \dfrac{1}{x}$. Write $(f \circ g)(x)$.

Solution: Substitute and clear the compound fraction by multiplying top and bottom by x:

$$f\left(\tfrac{1}{x}\right) = \frac{2}{\frac{1}{x} - 1} = \frac{2x}{1 - x}$$

$$\boxed{(f \circ g)(x) = \frac{2x}{1 - x}}$$

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6. $f(x) = \sqrt{x + 4}$, $g(x) = 2x - 1$. Solve $(f \circ g)(x) = 5$.

Solution: $(f \circ g)(x) = \sqrt{(2x - 1) + 4} = \sqrt{2x + 3}$, so

$$\begin{aligned} \sqrt{2x + 3} &= 5 \\ 2x + 3 &= 25 \\ 2x &= 22 \end{aligned}$$

Check: $\sqrt{2(11) + 3} = \sqrt{25} = 5$ ✓. $\boxed{x = 11}$

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7. $f(x) = 3x - 2$ and g piecewise ($-x$ for $x \leq 0$; x^2 for $x > 0$). **(a)** $(f \circ g)(-5)$. **(b)** $(g \circ f)(-5)$.

Solution (a): $-5 \leq 0$, so $g(-5) = -(-5) = 5$; then $f(5) = 3(5) - 2 = 13$.

$\boxed{(f \circ g)(-5) = 13}$

Solution (b): Now f runs first: $f(-5) = 3(-5) - 2 = -17$. Since $-17 \leq 0$, $g(-17) = -(-17) = 17$. The two answers differ, which is the point — composition is not commutative.

$\boxed{(g \circ f)(-5) = 17}$

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8. $f(x) = x^2 - 6x$, $g(x) = x + 3$. Write $(f \circ g)(x)$.

Solution: Replace each x in f by $x + 3$:

$$(x + 3)^2 - 6(x + 3) = x^2 + 6x + 9 - 6x - 18$$

The linear terms cancel. $\boxed{(f \circ g)(x) = x^2 - 9}$

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9. $f(x) = x^2 + 2x$, $g(x) = x - 3$. Solve $(f \circ g)(x) = 0$.
Solution: Do not expand — the composed form factors immediately.

$$\begin{aligned} (x-3)^2 + 2(x-3) &= 0 \\ (x-3)\big[(x-3) + 2\big] &= 0 \\ (x-3)(x-1) &= 0 \end{aligned}$$

$$\boxed{x = 1 \quad \text{or} \quad x = 3}$$

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10. f piecewise ($x + 7$ for $x < 2$; $3x$ for $x \geq 2$). **(a)** $(f \circ f)(-1)$. **(b)** Why the two passes differ.

Solution (a): First pass: $-1 < 2$, so $f(-1) = -1 + 7 = 6$. Second pass: the input is now 6, and $6 \geq 2$, so $f(6) = 3(6) = 18$. $\boxed{(f \circ f)(-1) = 18}$

Solution (b): *Instructor-judged.* A correct response tracks the two passes separately and says that the branch is decided by whatever number is being fed in at that moment: -1 selects $x + 7$, and the output 6 then selects $3x$. An answer that applies one rule twice — $(-1 + 7) + 7 = 13$, or $3(-1)$ then $3(-3)$ — has treated the branch as a property of the problem instead of a property of the input.

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11. $f(x) = \dfrac{x}{x+1}$, $g(x) = \dfrac{1}{x-2}$. Write $(f \circ g)(x)$.
Solution: Substitute, then multiply numerator and denominator by $x - 2$:

$$f\left(\tfrac{1}{x-2}\right) = \frac{\frac{1}{x-2}}{\frac{1}{x-2} + 1} = \frac{1}{1 + (x-2)}$$

$$\boxed{(f \circ g)(x) = \frac{1}{x-1}}$$

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12. $f(x) = \sqrt{x}$, $g(x) = x^2 - 3x$. **(a)** Solve $(f \circ g)(x) = 2$. **(b)** Why neither answer is extraneous.

Solution (a): $(f \circ g)(x) = \sqrt{x^2 - 3x}$, so

$$\begin{aligned} \sqrt{x^2 - 3x} &= 2 \\ x^2 - 3x &= 4 \\ x^2 - 3x - 4 &= 0 \\ (x-4)(x+1) &= 0 \end{aligned}$$

$$\boxed{x = -1 \quad \text{or} \quad x = 4}$$

Solution (b): *Instructor-judged.* The test is the radicand, not the sign of x. At $x = -1$: $1 + 3 = 4 \geq 0$; at $x = 4$: $16 - 12 = 4 \geq 0$. Both inputs leave a non-negative quantity under the square root, and both give $\sqrt{4} = 2$, so neither is extraneous. Rejecting -1 because it is negative confuses the sign of the input with the sign of the radicand.